Mohammed Abdul Muqeeth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.96
NARAYANA ,FILM NAGAR	Class XII (TGBIE)	2023	96.90%
NARAYANA CONCEPT SCHOOL	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

- Achieved a peak rating of **1656,** hitting **Expert** on Codeforces: muqeeth26832
- Secured **Global Rank 279** in Codeforces 1116(Div.2)
- Represented IIT Hyderabad 2X at **Inter-IIT Tech Meet** and secured **highest score** in the Qtrinos problem statement.
- Selected for **Google Upskilling Program** after clearing two DSA screening rounds.
- Solved **1000+ DSA problems** across Codeforces, LeetCode and CSES.

WORK EXPERIENCE

26MilesCapital

Quantitative Developer Intern

- Optimized a low-latency **C++17** execution engine, through active-instrument partitioning, cache-aware data layouts, and allocation optimizations, , reducing critical-path overhead in latency-sensitive trading workflows.
- **Improved cache locality and memory efficiency** via hot/cold data separation, instrument-ID based state management, and elimination of string copies in latency-sensitive code paths; benchmarked performance under CPU-core isolation.
- Built a **Python analytics framework** generating execution-quality metrics, fill statistics, markouts, Greek-PnL attribution, exposure analytics, and reproducible cross-version simulation reports.
- **Redesigned the execution framework** improving maintainability, extensibility, and deployment workflows; adopted for execution research and **partially deployed to production**.
- Developed **production risk controls** enabling dynamic exposure reduction, liquidation, and strategy halts without disrupting live trading systems. Currently **deployed in production**.

GIAN (Global Initiative of Academic Networks)

Full Stack Developer

- Built scalable REST APIs using **Go (Gin)** and **PostgreSQL** to support real-time application workflows.
- Managed **production deployments** and cloud infrastructure on **AWS (EC2, S3), Docker, Nginx, and Caddy**, handling service configuration, monitoring, and rollouts.
- Diagnosed and resolved production issues through log analysis and root-cause **debugging**, including fixing a critical PDF upload failure caused by a third-party library incompatibility.

KEY PROJECTS

Order Book Exchange Simulator (GitHub)

C++20 · STL· *Multithreading* · *OOP* · *Google Test*

- Built a **low latency matching engine in C++20** with price-time priority, supporting GTC, Market, FOK, FAK, GFD, Cancel, and Modify orders using std::map price levels and std::list FIFO queues.
- Designed concurrent order lifecycle management using **std::mutex, condition_variable, and atomic**; achieved **O(1) order lookup and cancellation** via unordered_map with stable list iterators.

- **Reduced lock contention** through batch locking and developed an ANSI terminal UI with live order-book depth, spread, and VWAP metrics; validated matching correctness using **Google Test**.

LowLatencyCPP *(GitHub)*

C++20 · Linux · Concurrency

- Built a header-only **C++20 low-latency library** featuring **Arena, Slab, and MemoryPool allocators**, a lock-free **SPSC queue**, and **TAS, TTAS, CLH, and MCS locks**; benchmarked performance using **Google Benchmark**.
- Reduced allocation latency from **~85 ns to ~16 ns (5.2× faster than malloc)** using a bump-pointer Arena allocator; achieved **5.8B transfers/sec** with a cache-line-padded SPSC ring buffer (**24× faster than a mutex-based queue**).
- Benchmarked **10 active-set data structure designs** across 6 workload patterns, achieving **30× faster iteration** and a **7.7× scheduler speedup;** analyzed lock contention across TAS,TTAS and futex-based std::mutex.

Emergency Queue System (GitHub)

Go · React · TypeScript · MongoDB· Docker

- Built a **real-time hospital queue management platform** with web and mobile interfaces supporting **priority-based patient triage** and live queue tracking.
- Designed and deployed a scalable **Go backend** with real-time queue updates using **PusherJS**, backed by a **Dockerized infrastructure** for reliable deployment.
- Implemented **fault-tolerant queue workflows** and **role-based access control** for hospital staff and administrators, ensuring secure and efficient queue management.

TECHNICAL SKILLS

- **Languages:** C++ (17,20) , C, Python, Go, JS,TS.
- **Systems & Performance:** Linux, Multithreading, Concurrency, Cache-Aware Programming, STL Internals, TCP/IP,Socket Programming,Profiling & Debugging (perf, gdb, Google Benchmark, Google Test)
- **Backed & Infra:** Git, Docker, AWS (EC2, S3), SSH, SCP,Tmux, REST APIs, PostgreSQL, MySQL, MongoDB, Nginx
- **Frameworks & Libraries:** Go (Gin, Fiber), FastAPI, Node.js (Express), React, Astro, Numpy, Pandas, Matplotlib,CMake,Make

POR

- **Head, Dev Club | 2025- 2026:** Led the development team in building and maintaining club platforms, mentoring junior developers, and conducting technical sessions to strengthen software development practices among students.

- **OCS Tech Cell | '25-'26:** Designed and deployed a credit requirement tracking platform by owning database schema design, backend API development, and dashboard implementation, enabling automated eligibility analysis for recruiters.

RELEVANT COURSES

- **Computer Science:** Programming , Data Structures , Algorithms, Parallel and Concurrent Programming , Operating System , Computer Architecture, Computer Networks, DBMS , Software Engineering , Data mining,Machine Learning, Deep Learning
- **Mathematics:** Discrete Math, Probability & Random Variables, Statistics, Linear Algebra, Calculus